MILITARY SPECIFICATION
SYSTEM, LOCAL CONTROL, M16A1E1

1. SCOPE

1.1 This specification *covers* a local control system, which consists of: Set, Wiring MIOAIEl, Controller, Drive, M12E2, and Gear, Oil, M6A1E1, used with Gun, Twin 40mm, Self-propelled, M42.

2. APPLICAbLE DOCUMENTS

2.1 The following specifications and drawings of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-G-793	Controller, Drive, M12Al and M12E2
MIL-G-13507	Gear, Oil, M6A1E1
MIL-S-13544	Set, **Wiring, MIOAIEl**

U. S. Army

51-7	Instruments, Fire Control; General Specifications Governing the Manufacture and Inspection of

DRAWINGS

Ordnance Corps

F7665876	System, Local Control, M16AlBl

(copies of **specifications, standards,** drawings, and publications. required by contractors in connection with specific procurement functions should be obtained from the procuring **activity or as directed by the contracting officer).**

3. REQUIREMENTS

3*1 General.- Unless otherwise specified, the requirements common to all fire control instruments as specified in Specification 51-7 shall be **mandatory.**

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3.2 Material, - Material shall be in accordance with the applicable drawings, and specifications forming a part of this specification.

3.3 Voltage inPut.- The local control system shall meet the perfonnance requirements of this specification when 26.5 volts direct current is applied to the input of the contact ring.

3.4 Performance

3.4.1 Drive controller.- The drive controller shall meet the requirements of Specification MIL-C-793.

3eke2 Wiring set.- The wiring set shall meet the requirements of Specification MIL-S-13544.

3.4.3 Oil gear.- The oil gear shall meet the requirements of Specification MIL-G-13507.

304.4 Power switch.- When the on turret fire control system power switch is in the IN position, the control selector switch in the LOCAL position, and the elevation and azimuth interlock s-witches are closed, the oil gear slewing clutches shall engage smoothly and with a minimum engagement of 3/32 inch, the elevation oil gear motor shall start immediately. The azimuth oil gear motor and the inverter then shall start in successive order. When the control selector switch is moved to the REMOTE position, the inverter shall cease operating.

3*4*5 Firing triggers- Firing solenoids shall fire the gun when either or both firing triggers switches On the drive controller are depressed and the safety switch oz the drive controller is in the ON position. (On systems utilizing a means of off carriage control of firing, a shorting type plug shall be connected to the training receptacle).

3.4.6 Signal lights.- When off carriage fire control is being used, the signal lights shall be energized when the control switch on the carriage is in the REMOTE position and deenergized when in the LOCAL position.

3.4.7 Slewing orientation.- When the power switch and the drive controller rate switch are in the ON position, and the drive controller handgrips are rotated in a clockwise direction (as viewed from above), the gun mount shall rotate clockwise in azimuth Counterclockwise rotation of handgrips shall cause the gun mount to rotate counterclockwise in azimuth. When the lower ends of the handgrips are pushed forward the guns shall elevate; when they are pulled back the guns shall depress.

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3.4.8 Slewing speeds.- With the lim s y s t e m properly, the gun mount on a 10 degree, and the handgrips of the drive controller displaced to their maximum limits, the following minimum velocities shall be obtained:

3.4.8.1 Azimuth.- Azimuth rate shall be 40 degrees per second minimum f clockwise and counterclockwise rotations. The difference between the velocity readings in each direction shall not exceed 3 degrees per second.

3.4.8.2 Elevation and depression,- Elevation and depression rates **shall be 25 degrees per second minimun.** The **difference between the velocity readings shall not exceed 3 degrees per second.**

3.4.9 Surge, - At constant rate (drive controller handgrip locked) the change of **rate shall not exceed 1.5 roils** of gun **motion when measured in accordance with 4.2.2.**

3.4.10 Hydraulic limit stoPs.- The guns shall be brought to rest smoothly when slewed into either elevation or depression limit stop at the maximum slewing speed. The stops shall be adjusted to minus 3 ± 0.5 degrees in depression and 85 ± 1.0 degrees in elevation.

3.4.11 Creepage.- When all components have reached their normal operating condition and the drive controller rate motor switch is in **the** OFF position, there shall be no creepage of the gun. When the rate motor switch is in the ON position and the handgrips of the drive controller are displaced 3 to 5 degrees from their neutral position and then released, the gun velocities shall be within 0.6 degrees per second in both azimuth and elevation. These rates shall be checked ever a period of one minute.

3.5 **Shock** and vibration.- The local control systems shall not suffer damage or subsequently fail to meet the requirements of this specification after they have been subjected to the road test and proof firing test specified in 4.3.2.

3.6 Workmanship.- Workmanship shall be in accordance with Specification 51-7.

4. QUalITY. ASSURANCE PROVISIONS

4.1 General.- Unless otherwise specified, each local control system shall be inspected for conformance with this specification and Specification 51-7. Unless otherwise specified, standard ambient tests shall be performed at a temperature of 60°F to 90°F, at a barometric pressure of 28 to 32 inches of mercury, and a relative humidity not greater than 80 percent.

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4.2 **Performance.-** To test the local control system for conformance with the performance requirements of this specificatior.~ it shall be mounted on its appropriate gun mount and carriage. The automatic loaders of each gun shall contain weight equivalent **to two** clips of ammunition: weight of the operating crew shall be simulated **with suitably distributed mass of approximately** 650 **pounds carried** in the turret.

4.2.1 Slewing sPeeds.- Slewing **speeds shall be tested as** follows:

4.2.1.l Azimuth

(a) Place carriage on a 10 degree incline and set guns at **zero elevation.**

(b) **Slew guns in azimuth until the maximum** velocity of **rotation is** obtained. Record that value.

(c) **Repeat step (b) for opposite direction of rotation.**

(d) Difference in *velocity* **for both directions** shall be **no greater than 3 degrees per second.**

4.2.1.2 Elevation and depression

(a) place Carriage on **a** 10 **degree incline and** slew **guns in azimuth** until the trunnion cant is 10 degrees.

(b) Elevate guns from maximum limit in depression and measure Elevation velocity between plus 20 ± and plus 80 degrees.

(c) Depress guns from maximum **limit in elevation and measure the depression velocity between plus** 60 degrees and 0 degrees.

(d) Difference in velocity for both directions shall be no **greater than 3 degrees per second.**

4.2.2 **Surge.- The change** of rate shall be measured on the gun mount by matching the rate from a constant speed motor against the rate of-the gun mount. These rates are the inputs to a differential; the output of the differential will then be the change in rate, Rates shall be measured at various gun mount speeds between 200 and 500 **mils/sec.**

4.3 **Environmental tests**

4.3.1 General.- The environmental tests specified shall be limited to those contained in the following paragraph. The local control

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system shall be assembled to its gun mount and carriage for these tests. Three of the first five systems and one out of every additional 100 or fraction thereof shall be tested.

4.3.2 **Shock and vibration.** The **local control system** shall be assembled to its gun mount **and carriage and driven for a distance of at least 50 miles over roads and under conditions simulating as closely as possible those found in actual service. Travel on hard surface roads shall include speeds up to and including 40 mph, travel on** unimproved **cross-country** roads shall include speeds up to **and including 20 mph. This test may be run concurrently** with the test called for in the vehicle specifications. **Proof firing shall be as designated in the gun mount specification.**

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with instructions forming **a part of** the **conract or** furnished by the Contracting officer in **connection with the contract/**

6. NOTES

6.1 **Ordering data. - Contract and invitations for bids should specify** the **title, number, date of this specification, and type of packing.**

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either in procurement or [illegible] in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that adequate products can be procured with a minimum amount of delay and at the least cost. Comments on the return of this sheet will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEM PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and address) | DATE

DD [illegible] 1426 [illegible]

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